

Internal Gravity Wave (IGW) Modeling

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1 Motivation 1 - displace a parcel

Displace a parcel upward adiabatically in some background fluid and see what happens. This is probably solvable in the Boussinesq approximation or something slightly more complex.

Let's start with all the equations and see where to go. The codes and references are stored here: The relevant codes are at <https://astronomy.nmsu.edu/jasonj/IGW/>.

1.1 Fluid equations

Follow Vallis [2017] 2.4.2. We will just consider the (x, z) plane, with horizontal and vertical motions (u, w) by convention. Let the density be composed of a constant value plus small perturbations:

$$\rho = \rho_0 + \delta\rho(x, z, t). \quad (1)$$

(If ρ_0 is not a function of z , then a displaced blob won't be denser than its surroundings, so later on I add a linear term.)

The pressure is similar, except the background is stratified:

$$p = p_0(z) + \delta p(x, z, t). \quad (2)$$

The background pressure is in hydrostatic equilibrium (fluctuations don't have to be):

$$\frac{dp_0}{dz} = -\rho_0 g. \quad (3)$$

1.1.1 Momentum equation

In the momentum equation, ignore rotation and consider an inviscid fluid. Then

$$\rho \frac{D\mathbf{v}}{Dt} = \rho \frac{\partial \mathbf{v}}{\partial t} + \rho(\mathbf{v} \cdot \nabla)\mathbf{v} = -\nabla p + \rho \mathbf{g}. \quad (4)$$

Using the expansions above, to first order (velocity-velocity terms are 2nd order) this becomes

$$\frac{\partial \mathbf{v}}{\partial t} = -\frac{1}{\rho_0} \nabla \delta p - \frac{\delta \rho}{\rho_0} \mathbf{g}. \quad (5)$$

As in Vallis [2017], let $\phi = \delta p / \rho_0$ (deviation pressure) and $b = -g\delta\rho/\rho_0$ (buoyancy) to get

$$\frac{\partial \mathbf{v}}{\partial t} = -\nabla\phi + b\hat{k}, \quad (6)$$

noting that ρ_0 can be included in the gradient since it's constant. This is Vallis [2017] 2.94 for our case.

Now, Vallis [2017] goes on to say that it's possible for the fluctuations to be in near hydrostatic equilibrium

$$\frac{d\phi}{dz} = b, \quad (7)$$

if the vertical acceleration w' is small, obviously, compared to the buoyancy term.

In any case, the two components of Eq. (6) are

$$\frac{du}{dt} = -\frac{d\phi}{dx}, \quad (8)$$

$$\frac{dw}{dt} = -\frac{d\phi}{dz} - b. \quad (9)$$

1.2 Continuity equation

The general equation

$$\frac{D\rho}{Dt} + \rho\nabla \cdot \mathbf{v} = 0 \quad (10)$$

becomes at first order

$$\frac{D\delta\rho}{Dt} + (\rho_0 + \delta\rho)\nabla \cdot \mathbf{v} = 0. \quad (11)$$

Using some arguments I don't quite understand that appeared first in Vallis [2017] 1.9.2:

The fact that $\text{div } \mathbf{v}=0$ does not imply that we may independently apply $D\rho/Dt=0$.

and

Furthermore, incompressibility does not necessarily imply the neglect of density variations in the momentum equation; it is only in the mass continuity equation that density variations are neglected ...

So we will throw away b terms here but not the ones above in the momentum equation. Thus we have

$$\nabla \cdot \mathbf{v} = 0, \quad (12)$$

or

$$\frac{\partial u}{\partial x} + \frac{\partial w}{\partial z} = 0. \quad (13)$$

1.2.1 Equation of state

So, $D\rho/Dt$ being nonzero comes into play in the thermodynamics. More to come here. See Vallis [2017] 2.4.2.

1.3 Boussinesq approximation

So now how do we arrive at a simplified problem that can be solved?

An important statement is on page 84 in Vallis [2017]:

Thus, for constant density fluids the gravitational term has no dynamical effect: there is no buoyancy force, and the pressure term in the horizontal momentum equations can be replaced by p' .

An equation like 2.106 really is $D \rightarrow d$ and there is no advection term. Vallis [2017] Section 7.3 is important for us.

1.3.1 Formulation 1

Eq. (7.57) of Vallis [2017] is what I coded first. Ignore pressure fluctuations (horizontal motions). Then, the equations to solve are

$$\begin{aligned}\frac{dw}{dt} &= b, \\ \frac{db}{dt} &= -N^2 w.\end{aligned}\tag{14}$$

The first equation comes from the vertical momentum equation and ignoring pressure fluctuations altogether. The second equation comes from a thermodynamic relation I haven't written here yet.

In any case, Equations (14) are trivial to solve analytically for w and b by making a 2nd-order equation. The solution gives the amplitude of w or b varying at frequency $\omega = N$. But if we start with a perturbed initial condition, does it vary in position as well?

1.3.2 Formulation 2

Sutherland [2010] Section 3.3. and 3.7 has non-Boussinesq problem.

Coming soon ... coded but not written down yet.

1.4 Brunt Väisälä frequency

But should $N = N(z)$? If I write $\rho = \rho_0 + \bar{\rho}(z) + \rho'(x, z, t)$ where $\bar{\rho}(z)$ is linear in depth, then N is still a constant.

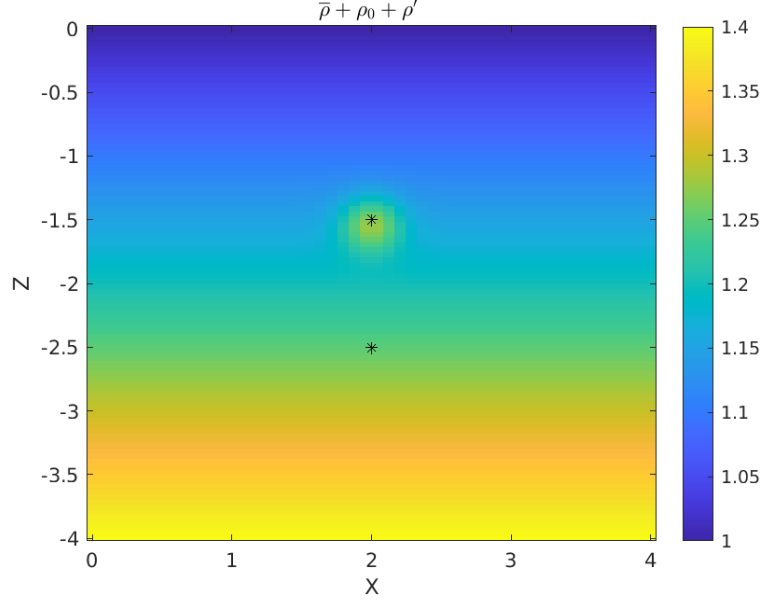


Figure 1: Background density and initial condition for $b \sim \rho'$ for the simple Boussinesq model. The color scale represents the density stratification and the top black asterisk is the location of the initial density perturbation (taken from the area centered on the lower asterisk).

$$N^2 = -\frac{g}{\rho_0} \frac{d\bar{\rho}}{dz}. \quad (15)$$

1.5 Numerical implementation

The relevant codes are at <https://astronomy.nmsu.edu/jasonj/IGW/codes/>.

Equations (14) are solved in the code `boussinesq.m`. I use a density expansion like given above in Section 1.4. I choose $\rho_0 = 1$. Then

$$\bar{\rho}(z) + \rho_0 = -0.1z + 1, \quad (16)$$

where $z = (-4, 0)$ in arbitrary units. The background density $\rho_0 + \bar{\rho}(z)$ is shown in Figure 1.

The density scale height for this case

$$H_\rho = -\left(\frac{d \log \bar{\rho}}{dz}\right)^{-1} \quad (17)$$

ranges from 10 to 14 length units, which is much larger than the scale of the domain. Hence, I think, the Boussinesq condition is satisfied. Taking $g = 50$ to increase the B.V. frequency and shorten the period, we have $N^2 = 5$ and $P = 2\pi/N = 2.81$. Taking a time domain of $t = (0, 10)$ allows us to see several periods.

For the initial conditions, I take a small gaussian ball of the background density (centered at around $z = -2.5$ and displace it vertically by some amount of vertical distance units (about 25 in this case). It is also shown in Figure 1, where the ball of greater density is location at about $x = 2$

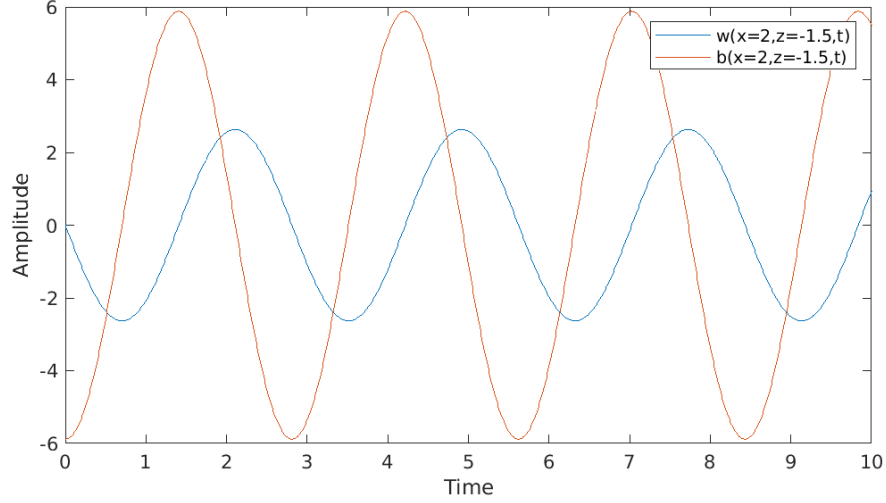


Figure 2: Amplitude of w and b at the center of the perturbation (around $z \approx -1.5$) as a function of time. The period is determined by N .

and $z = -1.5$, noted by the upper black asterisk. The original depth is the lower black asterisk. This is the initial condition for $b(x, z, t = 0)$ at the first time step. The initial condition for the vertical velocity w is $w(x, z, t = 0) = 0$.

The equations are solved using a simple 4th-order Runge-Kutta from $t = 0$ to $t = 10$ with $dt = 0.01$. This is in a script that's called and named `boussinesq_simple_rk4.m`.

The amplitude at the center of the perturbation of each variable as a function of time is shown in Figure 2. The two quantities vary at the given period, but are out of phase by a quarter of a period, or $\pi/2$. (Have to check if this is the expectation ...)

The animated gif of the time series of $w(x, z, t)$ and $b(x, z, t)$ is here: <https://astronomy.nmsu.edu/jasonj/IGW/FIGURES/>.

The filename is `bouss_simple.gif`. The panels are w , b , and $\rho' = -b\rho_0/g$.

1.5.1 Interpretation

The amplitude varies with the right period, but the density perturbation (parcel) does not oscillate about the equilibrium point. I want to think that it should be bobbing up and down around the equilibrium position, which is from where I took the density perturbation ($z = -2.5$) before displacing it to $z = -1.5$. Since the vertical speed w is finite and goes from positive to negative, I want to interpret that as the parcel accelerating about an equilibrium point. But it doesn't visually change in position.

Indeed, if I look at how much the parcel should be moving by computing $z \approx \int w(t) dt$ over the positive amplitude part of one period, it matches the distance over which I displaced the parcel to make the initial conditions. So the parcel *is moving*, it's just not moving.

2 Motivation 2 - reproduce water tank experiment

I want to numerically reproduce this experiment: https://www.youtube.com/watch?v=BDQD_gM3M24. This was carried out by Peter Rhines at the University of Washington.¹

To do this, we can (but maybe don't have to?) go beyond Boussinesq, and we also need a periodic forcing function to replicate the rod.

I've coded this too but haven't written it down yet except on a few pages. Coming soon ...

¹<https://www.ocean.washington.edu/research/gfd/>

References

B. R. Sutherland. Internal Gravity Waves. Cambridge University Press, 2010.

G. K. Vallis. Atmospheric and Oceanic Fluid Dynamics: Fundamentals and Large-Scale Circulation. Cambridge University Press, 2 edition, 2017. doi: 10.1017/9781107588417.